线性代数第三次作业

2 (1)
$$\begin{pmatrix}4&3&1\\1&-2&3\\5&7&0\end{pmatrix}\begin{pmatrix}7\\2\\1\end{pmatrix}=\begin{pmatrix}4\times7+3\times2+1\\7+2\times(-2)+3\\35+14+0\end{pmatrix}=\begin{pmatrix}35\\6\\49\end{pmatrix}$$

(2)
$$\begin{pmatrix}1&2&3\\-2&1&2\end{pmatrix}\begin{pmatrix}1&2&0\\0&1&1\\3&0&-1\end{pmatrix}=\begin{pmatrix}1+9&2+2&2-3\\-2+6&-4+1&1-2\end{pmatrix}=\begin{pmatrix}10&4&-1\\4&-3&-1\end{pmatrix}$$

(3)
$$(1\ 2\ 3)\begin{pmatrix}3\\1\\2\end{pmatrix}=3+2+6=11$$

(4)
$$\begin{pmatrix}2\\1\\3\end{pmatrix}(-1\ \ 2)=\begin{pmatrix}-2&4\\-1&2\\-3&6\end{pmatrix}$$

(5)
$$\begin{pmatrix}b_1&&&\\&b_2&&\\&&\ddots&\\&&&b_n\end{pmatrix}\cdot\begin{pmatrix}a_{11}&a_{12}&\cdots&a_{1n}\\a_{21}&a_{22}&\cdots&a_{2n}\\\vdots&\vdots&\ddots&\vdots\\a_{m1}&a_{m2}&\cdots&a_{mn}\end{pmatrix}=\begin{pmatrix}a_{11}b_1&a_{12}b_1&\cdots&a_{1n}b_1\\a_{21}b_2&a_{22}b_2&\cdots&a_{2n}b_2\\\vdots&&\ddots&\vdots\\a_{m1}b_n&a_{m2}b_n&\cdots&a_{mn}b_n\end{pmatrix}$$

(6)
$$(x_1\ x_2\ x_3)\begin{pmatrix}a_{11}&a_{12}&a_{13}\\a_{12}&a_{22}&a_{23}\\a_{13}&a_{23}&a_{33}\end{pmatrix}\begin{pmatrix}x_1\\x_2\\x_3\end{pmatrix}=(x_1\ x_2\ x_3)\begin{pmatrix}a_{11}x_1+a_{12}x_2+a_{13}x_3\\a_{12}x_1+a_{22}x_2+a_{23}x_3\\a_{13}x_1+a_{23}x_2+a_{33}x_3\end{pmatrix}$$
$$=a_{11}x_1^2+a_{12}x_2x_1+a_{13}x_1x_3+a_{12}x_1x_2+a_{22}x_2^2+a_{23}x_2x_3+a_{13}x_1x_3+a_{23}x_2x_3+a_{33}x_3^2$$
$$=a_{11}x_1^2+2a_{12}x_1x_2+2a_{13}x_1x_3+2a_{23}x_2x_3+a_{22}x_2^2+a_{33}x_3^2$$

5. (1) 解: $A=\begin{pmatrix}1&0\\1&0\end{pmatrix}$, $A^2=\begin{pmatrix}1&0\\1&0\end{pmatrix}\begin{pmatrix}1&0\\1&0\end{pmatrix}=\begin{pmatrix}1+0&0\\1+0&0\end{pmatrix}=\begin{pmatrix}1&0\\1&0\end{pmatrix}$

(2) 解: 令 $A=\begin{pmatrix}1&0\\0&0\end{pmatrix}$, $X=\begin{pmatrix}1\\1\end{pmatrix}$, $Y=\begin{pmatrix}1\\2\end{pmatrix}$

$AX=\begin{pmatrix}1\\0\end{pmatrix}$, $AY=\begin{pmatrix}1\\0\end{pmatrix}$, $AX=AY$

但 $X\neq Y$

7. ~~$B_{4\times4}$ 变换到 $A_{4\times4}$: $\alpha_1 \leftrightarrow \alpha_4$, $\alpha_1 \leftrightarrow \alpha_2$, $\alpha_1 \leftrightarrow \alpha_3$, $\alpha_3 \leftrightarrow \alpha_2$~~

~~共交换4次，故 $|B_{4\times4}| = |A_{4\times4}|$~~

$A+B = (\alpha_1+\alpha_3, 2\alpha_2, \alpha_3+\alpha_4, \alpha_4+\alpha_1)$

$$|A+B| \overset{c_1-c_4}{=\!=} |\alpha_3-\alpha_4, 2\alpha_2, \alpha_3+\alpha_4, \alpha_4+\alpha_1|$$

$$\overset{c_3-c_1}{=\!=} |\alpha_3-\alpha_4, 2\alpha_2, 2\alpha_4, \alpha_4+\alpha_1|$$

$$\overset{c_4-\frac{1}{2}c_3}{=\!=} |\alpha_3-\alpha_4, 2\alpha_2, 2\alpha_4, \alpha_1|$$

$$\overset{c_1+\frac{1}{2}c_3}{=\!=} |\alpha_3, 2\alpha_2, 2\alpha_4, \alpha_1|$$

$$= -|\alpha_1, 2\alpha_2, 2\alpha_4, \alpha_3| = |\alpha_1, 2\alpha_2, \alpha_3, 2\alpha_4|$$

$$= 4|\alpha_1, \alpha_2, \alpha_3, \alpha_4| = 4\times2 = 8$$

11. (1) 证明：$AB_1 = B_1A$, $AB_2 = B_2A$

$\therefore (B_1+B_2)A = B_1A+B_2A = AB_1+AB_2 = A(B_1+B_2)$, (B_1+B_2)与A可交换

$B_1B_2\cdot A = B_1AB_2 = AB_1B_2$, B_1B_2与A可交换

(2) 证明：$BA = AB$, $\therefore B^kA = B^{k-1}AB = B^{k-2}AB^2 = \cdots = A\cdot B^k$, B^k与A可交换

13 (1) 证明：$A = A^T$, $B = -B^T$

$(AB-BA)^T = (AB)^T-(BA)^T = B^TA^T - A^TB^T = (-B)\cdot A - A\cdot(-B) = AB-BA$, 为对称矩阵

$(AB+BA)^T = (AB)^T+(BA)^T = B^TA^T + A^TB^T = -BA-AB = -(AB+BA)$, 为反对称矩阵

(2) 证明：(充分性) $(AB)^T = B^TA^T = -B\cdot A$

$\because AB = BA$ $\therefore (AB)^T = -BA = -AB$, 为反对称矩阵

(必要性) $(AB)^T = -BA$, $-AB = (AB)^T$ (由反对称性)

$\therefore AB = BA$

$\therefore AB = BA \iff AB$为反对称矩阵

Ch2-1 作业：习题二：2,5,7,11,13

* 2 有些同学计算错误

2(3) 矩阵乘积的结果为 1 阶矩阵(11)，等价于一个数 11

2(6) 有少数同学题目看错，看成 $(x_1,x_2,x_3)\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$，实际应该是 $(x_1,x_2,x_3)\begin{pmatrix} a_{11} & a_{12} & a_{13} \\ a_{12} & a_{22} & a_{23} \\ a_{13} & a_{23} & a_{33} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}$. 另外还有同学计算结果没有合并同类项.

7 有些同学计算公式为 $|A+B|=|A|+|B|$错误，只能按一列的和进行拆分. 该题有多种解法，如下：

解： $|A+B|=|\alpha_1+\alpha_3,2\alpha_2,\alpha_3+\alpha_4,\alpha_4+\alpha_1|=\left|(\alpha_1,\alpha_2,\alpha_3,\alpha_4)\begin{pmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{pmatrix}\right|=|A|\begin{vmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 \end{vmatrix}=|A|\times 4=8.$

解法二： $|A+B|=|\alpha_1+\alpha_3,2\alpha_2,\alpha_3+\alpha_4,\alpha_4+\alpha_1|\overset{c_1+c_4-c_3}{=}|2\alpha_1,2\alpha_2,\alpha_3+\alpha_4,\alpha_4+\alpha_1|=4|\alpha_1,\alpha_2,\alpha_3+\alpha_4,\alpha_4|=4|A|=8.$

解法三：$|A+B|=|\alpha_1+\alpha_3, 2\alpha_2, \alpha_3+\alpha_4, \alpha_4+\alpha_1|=|\alpha_1, 2\alpha_2, \alpha_3+\alpha_4, \alpha_4+\alpha_1|+|\alpha_3, 2\alpha_2, \alpha_3+\alpha_4, \alpha_4+\alpha_1|$

$=|\alpha_1, 2\alpha_2, \alpha_3+\alpha_4,\alpha_4|+|\alpha_3, 2\alpha_2,\alpha_4, \alpha_4+\alpha_1|=|\alpha_1, 2\alpha_2, \alpha_3,\alpha_4|+|\alpha_3, 2\alpha_2,\alpha_4, \alpha_1|=2|A|+2|\alpha_1, \alpha_2, \alpha_3,\alpha_4|=8.$

13 有同学用元素来证明，相对繁琐一些，可用对称矩阵反对称矩阵的矩阵条件 $A^T=A$, $B^T=-B$ 来证，见如下：

证明：已知 $A^T=A$, $B^T=-B$.

(1) $(AB-BA)^T=(AB)^T-(BA)^T=B^TA^T-A^TB^T=-BA+AB=AB-BA$，故对称. 同理 $AB+BA$ 反对称.

(2) $(AB)^T=B^TA^T=-BA$，故有： AB 反对称⇔$(AB)^T= -AB$ ⇔$-BA= -AB$ ⇔ $AB=BA$.

13(2) 很多同学用充分性和必要性证明，也可用 “=>”和 “<=”表示证明方向，见如下：

证明：“=>” AB 反对称，则$(AB)^T= -AB$， 又有$(AB)^T=B^TA^T=-BA$，故$-AB =-BA$，即 $AB=BA$.

“<=” $AB=BA$，故$(AB)^T=B^TA^T=-BA= -AB$，即 AB 反对称 .

其余一些作业解题思路：

2 按矩阵乘法规则计算，并合并同类项，1 阶矩阵可以去掉矩阵括号

5 (1)可取 $A=\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$；(2) 可取 $A=\begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix},X=\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix},Y=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$，或者 $A=\begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix},X=\begin{pmatrix} 1 & 2 \\ 0 & 0 \end{pmatrix},Y=\begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$

11 (1) $(B_1+B_2)A=B_1A+B_2A=AB_1+AB_2=A(B_1+B_2)$； $(B_1B_2)A=B_1(B_2A)=B_1(AB_2)=(B_1A)B_2=(AB_1)B_2=A(B_1B_2)$；

(2) $BA^k=(B^{k-1}B)A=B^{k-1}(BA)=B^{k-1}(AB)=(B^{k-1}A)B=((B^{k-2}A)B)B=(B^{k-2}A)B^2=\ldots= AB^k$.